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Learning Anatomy in Disruptive Times: Physiotherapy Students' Perspectives on Blended Pedagogy in Higher Education

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Abstract

The COVID-19 pandemic necessitated the transition of physiotherapy programmes from traditional face-to-face anatomy teaching to fully online delivery. Studies conducted during this period have documented students' positive adaptation to online anatomy learning, highlighting benefits such as increased flexibility and reduced stress, while also noting challenges, including diminished practical exposure. Although much of the existing research centres on medical and dental education, there is a relative paucity of studies examining physiotherapy students' perspectives. In this study, a purposive sample of 53 entry-level physiotherapy students (Years 1–3) completed a questionnaire evaluating their experiences with anatomy education during the pandemic. Responses were compared between groups (Years 1 and 2 versus Year 3) using the Mann–Whitney U test and effect-size calculations. Year 3 students who experienced both face-to-face and online learning preferred practical sessions with silent mentors in the anatomy hall, citing greater active engagement and deeper learning. Conversely, all year groups reported that online assessments, such as the online Objective Structured Practical Examination, were less stressful than traditional formats. Year 1 and 2 students, lacking prior face-to-face practical experience, expressed neutral views regarding online practical components. Overall, while online theory and assessment components were well received, hands-on practical experience remains highly valued for promoting student engagement. These findings support the development of blended anatomy pedagogy that leverages the strengths of both modalities to enhance resilience and adaptability in the face of future educational disruptions.

Keywords: anatomy education; COVID-19 impact; physiotherapy education; online learning; blended learning

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1. Introduction

The COVID-19 pandemic has underscored the need for adaptability within global educational systems, particularly those that rely heavily on experiential, face-to-face pedagogies [1]. As the foundational science of medical and allied health curricula, anatomy education was uniquely challenged during this time due to its inherent emphasis on three-dimensional spatial understanding and hands-on experiences [2]. Traditionally, anatomy is taught through lectures, physical models, cadaveric dissections, and tutorials, all of which require physical presence and tactile interaction [3]. For centuries, cadaveric learning has

enabled students to engage deeply with human structures, fostering not only anatomical understanding but also the emotional and professional development of future healthcare providers [4,5]. In this context, the cadaver has often been regarded as a student's first patient, respectfully known as the silent mentor, and has facilitated the cultivation of empathy, situational awareness, and clinical professionalism.

Comprehensive anatomical knowledge is fundamental to clinical reasoning, assessment, and treatment planning for physiotherapy students across diverse patient presentations [6]. Although anatomy is a core subject in many health professions, its application in physiotherapy is notably intensive and hands-on. Physiotherapy education prioritises the functional and clinical integration of anatomical knowledge through physical assessment, manual therapy, and movement-based interventions [7]. Proficiency in musculoskeletal, nervous, and cardiorespiratory systems is essential for effective muscle strengthening, neuromuscular rehabilitation, and cardiorespiratory management. Students must grasp the detailed structures and functions of these systems to evaluate impairments, develop rehabilitation strategies, and implement interventions accurately. As a result, anatomy education underpins the ability to localise symptoms, identify red flags, and make differential diagnoses, all of which are critical for patient safety and optimal outcomes. Mastery of surface anatomy, palpation techniques, joint mechanics, and muscle actions bridges the gap between theoretical knowledge and clinical practice in physiotherapy [8]. Hands-on cadaveric interaction and engagement with physical models foster spatial reasoning and tactile familiarity, which are vital for building the confidence and competence required in clinical placements and future patient care [6]. Given its pivotal role in developing clinical competence, anatomy is not merely an introductory subject but a cornerstone supporting ongoing learning and skill integration throughout the physiotherapy curriculum [6]. Both students and educators recognise that a deep, applied understanding of anatomy enhances professional development, clinical reasoning, and lifelong learning in physiotherapy. Therefore, the quality and format of anatomy education are particularly influential in preparing physiotherapy graduates for professional practice. The mode of anatomy delivery, whether digital, face-to-face, or blended, has significant implications for students' readiness for clinical practice.

The abrupt transition to online learning during the pandemic necessitated a rapid transformation of anatomy teaching methods [9]. Digital platforms, including recorded demonstrations and 3D anatomy software, supported instructional continuity but could not fully replicate the depth of hands-on, spatial, and tactile learning. Although the literature increasingly explores advanced digital innovations in anatomy education, such as interactive 3D tools and virtual dissection systems [10,11], these technologies were not widely accessible or implemented across all institutions during the pandemic. Consequently, this study examines students' perceptions of the emergency remote adaptations that were feasible within the available time and resource constraints. Such adaptations also underscore the importance of resilience as a key concept in education, highlighting how students and educators navigate sudden disruptions. In this context, resilience refers to the capacity of educational systems, educators and students to adapt, maintain instructional effectiveness, and achieve learning outcomes despite disruption [12,13]. Academic resilience at the learner level has been described as the ability to cope with and continue progressing during academic challenges, such as the sudden transition to remote learning [14,15]. In the present study, resilience is therefore used to contextualise how students' experiences and preferences during the pandemic may inform future improvements to teaching delivery, rather than as an empirical variable directly assessed by the survey.

In addition to the shift to remote teaching, the pandemic led to the loss of hands-on practice and changes in assessment formats. Accordingly, this study had two complemen-

tary aims: (1) to evaluate physiotherapy students' perceptions of anatomy education during the COVID-19 pandemic, with a particular focus on comparing online and face-to-face learning; and (2) to identify student preferences regarding the delivery of anatomy learning materials during the transition to remote teaching. A further conceptual aim was to consider how these experiences and perceptions might inform the development of blended teaching models to enhance the adaptability of future anatomy curricula, without directly measuring academic resilience. As students were required to participate in remote learning rather than choosing it, their preferences were explored to determine which aspects of delivery they found supportive or limiting under constrained circumstances. These insights are valuable for informing the design of future blended teaching models, where student preferences can influence curricular decisions even if options were limited at the time.

This study was guided by the following research questions: (1) How did physiotherapy students perceive online anatomy education compared to traditional face-to-face teaching during the COVID-19 pandemic? (2) What were students' preferences regarding the delivery of anatomy teaching and assessment during the transition to remote learning? Insights from these descriptive findings are intended to inform potential strategies for enhancing the adaptability and future resilience of anatomy education, rather than to provide direct measurement.

2. Materials and Methods

2.1. Study Design

The design and reporting of this cross-sectional survey adhered to the Consensus-Based Checklist for Reporting of Survey Studies (CROSS) guidelines [16]. The University Institutional Review Board granted ethical approval (approval number 2022073). This purposive online recruitment survey collected data from April 2022 to February 2023 via QualtricsXM. All cohorts (Years 1 to 3) were surveyed concurrently during this period, and therefore, the timing of recruitment did not differ by year of study. Month-by-month response tracing was not undertaken because the survey was conducted as a single cross-sectional exercise rather than a staged or longitudinal project.

Data collection was anonymous, with the Anonymise Responses function enabled to prevent the retention of IP addresses and geolocation data. Participants provided informed consent electronically before beginning the survey. Participation was entirely voluntary, and no follow-up was required after completion. To minimise duplicate entries while maintaining anonymity, the Prevent Ballot Box Stuffing option was activated, restricting multiple submissions from the same device/browser. Only anonymised responses were used for data analysis to ensure participant confidentiality.

2.2. Participants

Participants were recruited from an entry-level physiotherapy programme at a local university. All Year 1, Year 2 and Year 3 physiotherapy students were invited to participate in the survey. Participants were recruited through cohort group emails and social communication platforms, with periodic reminders sent during the data collection to encourage participation and improve response rates.

Inclusion criteria were: (1) aged 18 years or older at the time of the survey, (2) currently enrolled in the physiotherapy programme, and (3) having completed at least one anatomy module during the COVID-19 pandemic period. Students were excluded if they (1) were under 18 years of age, (2) were no longer actively matriculated (e.g., withdrawn or on leave of absence), or (3) had completed all anatomy modules prior to the onset of the COVID-19 pandemic.

2.3. Sample Size

A target sample size of 79 participants was estimated from an overall student population of approximately 450, using standard sample size calculation principles [17]. Parameters were set at a 95% confidence level, a 10% margin of error, and an assumed response distribution of 50%, which provides a conservative estimate of the required sample size. These parameters were chosen to balance precision with feasibility, given the exploratory nature of the study and the unprecedented shift to online anatomy education during the COVID-19 pandemic.

2.4. Anatomy Education

Anatomy is a compulsory component of the first-year curriculum of all physiotherapy students. Delivered in conjunction with physiology, the subject is structured into two sequential modules, Anatomy and Physiology Part 1 and Part 2, across two academic terms. Each module is designed for 250 notional learning hours, including approximately 75 contact hours across lectures, tutorials, and practical sessions. Across all cohorts, the anatomy portion comprised approximately 16 h of lectures, 16 h of tutorials, and 20 h of practical sessions in the anatomy hall, with the remaining hours designated for independent study and assessment preparation. The curriculum content and topics remained consistent across cohorts, with differences primarily in the delivery approaches due to pandemic-related adaptations.

2.5. Before the COVID-19 Pandemic

Before the onset of the COVID-19 pandemic, anatomy was taught through traditional face-to-face methods, including lectures, tutorials, and practical sessions with silent mentors. Students typically attended lectures, tutorials, and face-to-face practicals in the anatomy hall. Groups of 6 to 8 students were assigned to each silent mentor to gain hands-on learning and direct engagement with real anatomical specimens. Faculty actively circulate during practical sessions within the anatomy hall to provide guidance and answer questions. Additionally, students had access to an anatomy museum with preserved specimens to reinforce their learning. This format was consistent across all cohorts up to 2020, providing Year 3 students (in 2022) with full face-to-face exposure.

2.6. During the COVID-19 Pandemic

During the COVID-19 pandemic (September 2020 to April 2023), anatomy education was rapidly transitioned to a fully online format. Lectures and tutorials were delivered synchronously via Zoom, following the same sequence and learning objectives as pre-pandemic teaching. Practical sessions were adapted by having faculty demonstrate anatomical structures on silent mentors in the anatomy hall, with students observing via live-streamed sessions. Recordings were provided for asynchronous review, and an online discussion forum supported ongoing clarification of content. Students retained access to the anatomy museum, subject to capacity restrictions and social distancing measures.

Although the curriculum content remained unchanged, the mode of delivery differed between cohorts:

- Year 3 (2019–2020): Experienced face-to-face lectures and practicals prior to the pandemic, transitioning to online practicals mid-term.
- Year 2 (2021): Began with fully online lectures and tutorials, with limited face-to-face practical access.
- Year 1 (2022): Experienced a combination of online lectures and tutorials with progressive resumption of face-to-face practical sessions.

2.7. Instruments

The survey was developed during the height of the COVID-19 pandemic, using a pragmatic, literature- and expert-informed approach, as no validated instruments were available at the time for assessing physiotherapy students' experiences of online anatomy education. The questionnaire was based on prior literature [18–26] and aligned with the aims of the present study. Parts 2 to 5 were directly derived from constructs reported in these studies, including perceived quality of online anatomy education, preferences for teaching modalities, adaptations to the objective structured practical examination (OSPE), and access to digital resources.

The questionnaire comprised five sections:

1. General demographic information, including age, sex, and year of study.
2. Seven questions were explicitly targeted at Year 3 students who experienced both face-to-face anatomy education (from September 2019 to February 2020) and online instruction (from February to April 2020) due to the pandemic-related shift. This section explored their personal preferences and comparative experiences across the two teaching modalities.
3. 14 items evaluating the quality of anatomy education during the COVID-19 pandemic.
4. 5 questions assessing students' experiences with the online OSPE, and
5. 3 questions related to students' accessibility to online resources during the pandemic.

Most items employed a 5-point Likert scale: Strongly disagree (1), Somewhat disagree (2), Neither agree nor disagree (3), Somewhat agree (4), and Strongly agree (5). Demographic questions and Question 30 (technical difficulties) did not use a Likert scale. The survey was administered in English, and Supplementary Information File S1 presents the survey questionnaire.

Survey Development and Validation

The survey was originally developed during the pandemic period, when validated instrument specific to physiotherapy anatomy education were not yet available. Three co-investigators drafted the initial items based on the available literature and study aims. Given the exploratory nature of this research and the limited availability of similar validated tools during the height of the COVID-19 pandemic, the survey was designed to address gaps in understanding students' experiences with anatomy education during this unprecedented shift.

The first 20 responses were examined as a pragmatic check of item clarity. No formal cognitive interviewing or psychometric validation (e.g., Cronbach's alpha, factor analysis) was performed. The instrument was used in its exploratory form because (1) the aim was to document perceptions rather than to test latent constructs; (2) resources and validated discipline-specific tools were limited at the time; and (3) the survey had already been deployed during the pandemic period, and retrospective validation is not feasible with the current dataset.

2.8. Data Analysis

All data were analysed using GraphPad Prism (version 8.4.3), with statistical significance set at $p < 0.05$. Descriptive statistics were used to summarise participants' responses, including frequencies and percentages for categorical variables and means, standard deviations (SD), and medians for ordinal or non-normally distributed data. The Mann–Whitney U test was used to compare responses from Year 1 and Year 2 students, analysed as a single group, with those from Year 3 students. Effect sizes were calculated using rank-biserial correlation (r) to complement p values and indicate the magnitude of

observed differences, where r ≈ 0.10 represents a small effect, r ≈ 0.30 a moderate effect, and r ≥ 0.50 a large effect.

Although sex was included in the demographic profile, it was not treated as an analytical variable, and no subgroup analyses were performed. Its inclusion served only to describe the sample characteristics.

3. Results

A total of 68 surveys were returned, representing a response rate of 15%. Fifteen datasets were excluded due to incompleteness; therefore, 53 fully completed responses (12%) were included in the final analysis. All incomplete questionnaires were excluded entirely, and no imputation procedures were applied. Because missing data were removed list-wise, it was not possible to determine whether missingness occurred at random. Although the final sample did not reach the calculated target size, the study was designed primarily as a descriptive exploratory survey, and findings are interpreted with appropriate caution regarding statistical power and the potential risk of Type II error.

Figure 1 depicts the flow of participant recruitment. Among all the valid datasets, 27 (11 males and 16 females) belonged to the Year 1 and Year 2 groups, and 26 (9 males and 17 females) belonged to Year 3. All participants were aged 20 to 36, with a mean of 24.11 (SD 3.3) years. Although sex was collected, subgroup analyses were not performed due to the small and uneven distribution of male (n = 20) and female (n = 33) students. These numbers were insufficient to support reliable statistical comparisons or regression modelling. Instead, sex was retained solely as a descriptive variable. For the same reason, age was not included as a covariate in inferential testing.

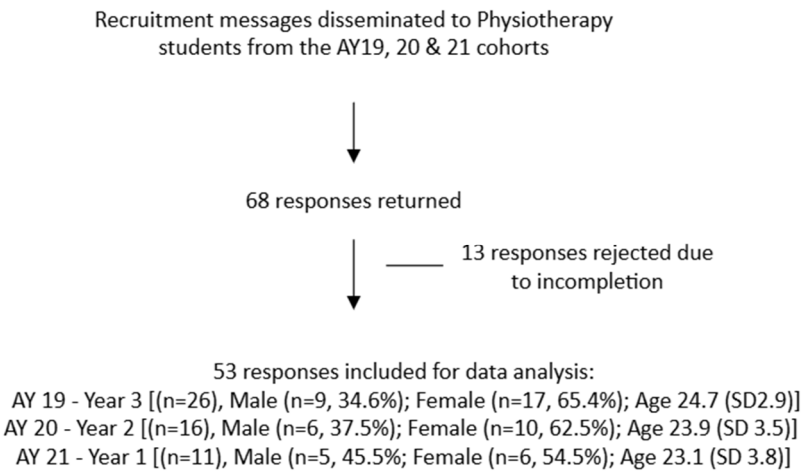

Figure 1. Participant recruitment flowchart.

Prior to analysis, the Mann–Whitney U test comparing Year 1–2 and Year 3 was designated as the primary planned comparison because these groups differed meaningfully in their exposure to face-to-face versus online anatomy instruction. All other between-group comparisons were considered exploratory. Given the study's descriptive aims, multiple-comparison corrections were not applied. However, results are interpreted cautiously, and the elevated risk of Type I error is acknowledged.

Table 1 presents the Year 3 participants' responses regarding their experiences and perceptions of face-to-face versus online anatomy education. Students reported that online teaching was less interactive than face-to-face sessions and preferred face-to-face interaction during tutorials and practicals. They also perceived the online OSPE as a less stressful alternative. These findings reflect students' subjective perceptions rather than

objective measures of learning effectiveness, but they highlight interaction as a key factor for consideration in designing future blended models.

Table 1. Score distribution when comparing Face-To-Face and online anatomy education.

	Question	Strongly Disagree n (%)	Somewhat Disagree n (%)	Neither Agree nor Disagree n (%)	Somewhat Agree n (%)	Strongly Agree n (%)	Total Mean (SD) Median
4	Online anatomy teaching was more effective than face-to-face lectures and tutorials.	1 (3.9)	17 (65.4)	4 (15.4)	4 (15.4)	0 (0.0)	2.4 (0.8) 2
5	Online anatomy education was as interactive as listening to the lecture face-to-face.	5 (8.3)	12 (46.2)	2 (7.7)	6 (23.1)	1 (3.9)	2.5 (1.2) 2
6	I missed the face-to-face interaction after the tutorials when I could clarify difficult concepts with the faculty.	0 (0.0)	1 (3.9)	1 (3.9)	12 (46.2)	12 (46.2)	4.4 (0.8) 4
7	The online anatomy lab sessions with the silent mentor were as effective as the face-to-face sessions at Anatomy Hall.	9 (34.6)	13 (50.0)	2 (7.7)	1 (3.9)	1 (3.9)	1.9 (1.0) 2
8	Overall, I would prefer attending online anatomy teaching than the face-to-face curriculum.	7 (26.9)	13 (50.0)	1 (3.9)	5 (8.3)	0 (0.0)	2.12 (1.1) 2
9	I felt less stressed during anatomy OSPE when it was conducted online compared to face-to-face.	2 (7.7)	2 (7.7)	3 (11.5)	12 (46.2)	7 (26.9)	3.8 (1.2) 4
10	I could focus better during online OSPE since I did not have to move between stations physically	2 (7.7)	4 (14.4)	3 (11.5)	11 (42.3)	6 (23.1)	3.6 (1.2) 4

Table 2 reports the evaluations and comparisons of the anatomy education that all participants experienced. Participants generally agreed that the online anatomy education is comparable to the face-to-face format, with most mean and median scores for Questions 11 to 26 ranging from 3 to 5, except for Question 16, which concerned interaction with classmates during Zoom lectures and tutorials. Remarkably, Questions 22, 23 and 26 were positively framed to reflect the face-to-face tutorial structure and practical lessons involving silent mentors in the anatomy hall. All cohorts consistently favoured this format, with mean and median scores exceeding 4 out of 5. Particularly for Question 23, "The face-to-face lab environment in the anatomy hall with the silent mentors was conducive to my learning." Year 1 and Year 2 students reported higher agreement (mean 4.9 ± 0.3; median 5) compared to Year 3 students (mean 4.4 ± 0.6; median 4), U = 186, $p = 0.004$ and almost significant effect size (r = 0.49). This effect size approaches the threshold for a large effect and is one of the more substantial differences observed across the dataset. Even where statistical significance was not achieved, effect sizes were examined to support interpretation, consistent with the exploratory nature of the study.

Table 2. Score distribution on quality of anatomy lesson during COVID-19.

	Question	Strongly Disagree n (%)	Somewhat Disagree n (%)	Neither Agree nor Disagree n (%)	Somewhat Agree n (%)	Strongly Agree n (%)	Total Mean (SD) Median	Year 1&2 Mean (SD) Median	Year 3 Mean (SD) Median	Mann–Whitney U	*p* Value	r_rb (*r*)
11	Anatomy education via online classes was clear and easy to follow	0	7	6	37	3	3.7 (0.8) 4	3.8 (0.9) 4	3.5 (0.7) 4	268	0.144	0.25
12	The teaching materials used in online education were sufficient for the purpose of the module, e.g., I did not have to source my own additional materials to supplement what was provided.	1	15	12	22	3	3.2 (1.0) 3	3.2 (1.1) 3	3.2 (0.9) 3	346	0.936	0.01
13	The duration of the online anatomy lectures was optimal to maintain concentration throughout.	4	9	14	24	2	3.2 (1.0) 3	3.1 (1.2) 3	3.3 (0.9) 4	318	0.565	0.09
14	I attended most of the online anatomy lectures during the live lecture time.	3	8	4	19	19	3.8 (1.2) 4	3.8 (1.4) 4	3.8 (1.1) 4	322	0.607	0.08
15	I prefer viewing most of the online anatomy lectures via the recordings for flexibility (play, pause, rewind, watch at faster/slower speed) and at my own time.	1	4	4	18	26	4.2 (1.0) 4	4.4 (0.9) 5	4.0 (1.1) 4	259	0.103	0.25
16	Interacting with my classmates during the online lectures and tutorials was easy via Zoom.	6	11	14	20	2	3.0 (1.1) 3	2.9 (1.2) 3	3.1 (1.0) 3.5	312	0.495	0.10
17	I was able to post questions easily to the instructor during the online teaching.	0	7	9	24	13	3.8 (1.0) 4	4.0 (0.9) 4	3.7 (1.0) 4	294	0.317	0.15
18	Question and answer sessions by instructors during online teaching had similar effectiveness as face-to-face lessons.	1	19	8	19	6	3.2 (1.1) 3	3.3 (1.2) 3	3.1 (1.0) 3	326	0.658	0.07
19	The online anatomy education was interactive and fun.	0	9	21	21	2	3.3 (0.8) 3	3.3 (0.8) 3	3.3 (0.78) 3	346	0.936	0.01

Table 2. *Cont.*

	Question	Strongly Disagree n (%)	Somewhat Disagree n (%)	Neither Agree nor Disagree n (%)	Somewhat Agree n (%)	Strongly Agree n (%)	Total Mean (SD) Median	Year 1&2 Mean (SD) Median	Year 3 Mean (SD) Median	Mann–Whitney U	*p* Value	r_rb (*r*)
20	The model visuals used in online E-museum sessions were sufficient to supplement what was taught during online lectures and tutorials.	2	12	11	24	4	3.3 (1.0) 4	3.3 (1.1) 4	3.3 (1.0) 3.5	344	0.908	0.02
21	The functional anatomy videos (e.g., anatomical structures used in shoulder rotation) used in online anatomy E-museum were sufficient.	1	8	15	27	2	3.4 (0.9) 4	3.4 (1.0) 4	3.4 (0.8) 4	342	0.880	0.02
22	Face-to-face anatomy tutorials were more effective for my learning.	1	4	4	19	25	4.2 (1.0) 4	4.1 (1.2) 5	4.2 (0.8) 4	325	0.645	0.07
23	The face-to-face lab environment in the anatomy hall with the silent mentors was conducive to my learning.	0	0	1	17	35	4.6 (0.5) 5	4.9 (0.3) 5	4.4 (0.6) 4	186	0.004 *	0.49
24	The 2-dimensional cadaver photos/diagrams during online E-museum were easy to understand and aided my learning by supplementing what was taught in lectures, tutorials and anatomy lab.	2	10	12	26	3	3.3 (1.0) 4	3.4(1.0) 4	3.3 (0.9) 3	321	0.601	0.08
25	The online E-museum was concise and focused	0	4	17	30	2	3.6 (0.7) 4	3.5(0.8) 4	3.7 (0.6) 4	310	0.473	0.11
26	Having silent mentors during face-to-face lab sessions made it easier to appreciate the relationship between structure and function, as well as anatomy landmarks.	0	0	2	6	45	4.8 (0.5) 5	4.9 (0.3) 5	4.7 (0.6) 5	294	0.117	0.22

Note: U = Mann–Whitney test statistic; p values are exact, two-tailed; r_rb = rank-biserial correlation (effect size); * denotes $p < 0.05$.

Six out of 53 participants (1 from Year 1, 2 from Year 2, and 3 from Year 3) reported technical problems, such as unstable internet connections and audio-visual issues, which contributed to frustration during online learning. While technical disruptions are not unique to the online environment, face-to-face instruction can similarly be affected by equipment malfunctions or logistical failures. The unfamiliarity and novelty of online-related issues during the pandemic may have amplified their impact on students' learning experiences, suggesting that future instructional design must account for technological resilience and support systems.

4. Discussion

This study examined physiotherapy students' perceptions of anatomy education during the pandemic, with particular attention to their experiences with both face-to-face and online modalities amid educational disruption. The objective was to derive insights that could inform the future development of resilient anatomy curricula.

The findings indicate that, although students recognised the efforts to maintain instruction through online delivery, they consistently preferred face-to-face lessons, especially for practical sessions. This preference is consistent with existing literature, which demonstrates that health professions students value hands-on experience with silent mentors in the anatomy hall. Such experiences are regarded as unparalleled in fostering spatial understanding, tactile learning, and emotional connection, qualities that digital tools, including virtual dissection platforms, 3D models, and interactive technologies, cannot fully replicate [10].

Within the broader digital anatomy literature, the present findings complement studies of advanced immersive technologies, including 3D virtual dissection platforms, virtual and augmented reality, and haptic systems [10,11]. These technologies have demonstrated the ability to enhance spatial visualisation and facilitate repeated, self-paced exploration of anatomical structures [27,28], particularly when access to cadaveric material is limited. However, they frequently abstract anatomy from its physical and social context. In contrast, the hybrid digital strategies examined in this study, live online instruction, recorded demonstrations, and virtual assessments, maintained curricular continuity while emphasising students' perceptions of interaction, stress, and learning authenticity. The results indicate that although low-immersion digital modalities may be limited in replicating tactile and spatial cues, they may better support affective dimensions such as perceived psychological safety and assessment-related stress. To situate the present findings, Figure 2 presents a descriptive conceptual framework that illustrates how teaching modality shapes learning affordances, which in turn influence students' perceptions and perceived adaptive outcomes during periods of disruption. Unlike studies evaluating advanced digital imaging, virtual dissection platforms, immersive VR/AR environments, or haptic systems, the present study does not assess the effectiveness of specific technologies. Instead, it captures how students perceive and respond to changes in learning affordances when traditional cadaveric exposure is constrained. The framework is heuristic and non-causal, intended to organise empirical findings and interpretations rather than to imply predictive or causal relationships.

Advanced digital modalities are designed primarily to enhance spatial visualisation, interactivity, and learner autonomy, particularly in the absence of physical specimens. In contrast, the emergency remote and blended strategies examined here foregrounded continuity, accessibility, and instructional adaptability rather than immersive simulation. Furthermore, students' strong preference for face-to-face practical sessions highlights the enduring value of embodied, socially situated learning, which immersive technologies alone may not fully replicate. Collectively, these observations indicate that advanced digital

tools and pragmatic online delivery address distinct dimensions of anatomy learning and should be regarded as complementary rather than substitutive.

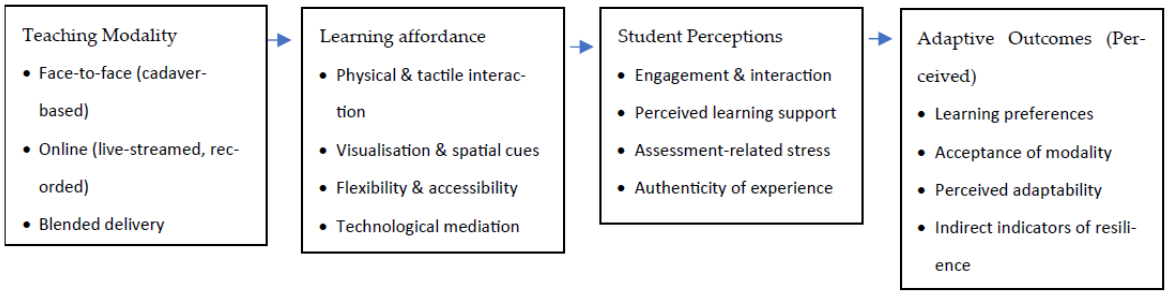

Figure 2. Descriptive framework illustrating factors shaping physiotherapy students' perception of anatomy education during pandemic-related disruption.

The Year 3 cohort, who uniquely experienced both face-to-face and online teaching, reported a generally neutral perception of the abrupt transition to online learning (Table 1). This neutrality aligns with studies indicating that, during crisis-driven transitions, students may prioritise continuity over quality and withhold judgment during periods of uncertainty [29,30]. Despite this, Year 3 students ultimately expressed a preference for face-to-face modalities, particularly for practical sessions, reflecting the broader consensus in anatomy education [4]. Subsequent cohorts (Years 1 and 2) demonstrated strong appreciation for their eventual return to the anatomy hall (Table 2, Question 23). As the pandemic progressed, institutions developed more effective strategies for managing campus operations, enabling a structured and cautious resumption of face-to-face lessons through measures such as cohort segregation, staggered scheduling, and stringent infection control protocols [31,32]. This facilitated students' re-engagement with hands-on anatomy education in a controlled environment, reinforcing their appreciation for face-to-face learning and the critical role of the anatomy hall in supporting anatomical understanding. Students consistently valued direct tactile learning experiences, underscoring the enduring importance of cadaveric learning and corroborating findings from the broader anatomy education literature [2,3,5].

A notable finding was that Year 3 students reported less stress during the online OSPE [Table 1, Question 9; mean 3.8 (SD 1.2), median 4], despite preferring face-to-face assessments. This aligns with well-established evidence that performance anxiety is reduced in remote or familiar environments [33,34]. However, students still favoured the authenticity and professional relevance of face-to-face OSPE conducted in the anatomy hall [Table 3, Question 29; mean 3.9 (SD 1.0), median 4]. In contrast, Years 1 and 2 students who had never experienced a traditional OSPE were less inclined to prefer face-to-face formats [Table 3, Question 29, mean 3.0 (SD 1.5); median 3]. Their perceptions likely reflect the lower pressure associated with completing assessments remotely. This pattern illustrates how familiarity with assessment shapes stress perception and should be considered when designing future blended assessment strategies [34].

Table 3. Score distribution on perception of online OSPE.

	Question	Strongly Disagree n (%)	Somewhat Disagree n (%)	Neither Agree nor Disagree n (%)	Somewhat Agree n (%)	Strongly Agree n (%)	Total Mean (SD) Median	Year 1 & 2 Mean (SD) Median	Year 3 Mean (SD) Median	Mann–Whitney U	*p* Value	r_rb (*r*)
27	The pictures shown during the online practical exam were a good representation of what I saw during the face-to-face lab sessions.	5	19	5	22	2	2.9 (1.2) 3	2.9 (1.2) 3	3.0 (1.1) 3.5	332	0.736	0.05
28	The markings used in the pictures during the online practical exam were readily identifiable.	2	19	9	18	5	3.1 (1.1) 3	3.2 (1.1) 3	3.0 (1.1) 3.0	308	0.451	0.11
29	I would have preferred OSPE to be conducted face-to-face in the anatomy hall than online.	7	6	12	14	14	3.4(1.4) 4	3.0 (1.5) 3	3.9 (1.0) 4	240	0.047	0.28
30	I prefer typing out my answers rather than writing them in an examination.	1	5	13	17	17	3.8 (1.1) 4	4.0 (1.2) 5	3.6 (0.9) 4	251	0.078	0.25
31	I find online OSPE difficult as I am not able to refer back to the previous questions (and had no rest stations unlike face-to-face OSPE).	2	5	11	22	13	3.7(1.1) 4	3.4 (1.1) 4	4.0 (0.9) 4	244	0.048	0.28

Note: U = Mann–Whitney test statistic; *p* values are exact, two-tailed; r_rb = rank-biserial correlation (effect size).

While cadaver-based anatomy courses are often justified by their role in fostering three-dimensional spatial understanding and, in some cases, empathy towards the human body, these outcomes are rarely measured directly at the undergraduate level. Our study similarly assessed perceptions rather than objective learning gains. As such, the findings should be interpreted as subjective evaluations rather than evidence of actual learning effectiveness.

The present findings aim to provide insight into academic resilience. Although resilience was not measured with a validated instrument, students' narratives reflect adaptive behaviours described in existing resilience frameworks, including persistence during rapid modality changes, utilisation of digital tools, and adjustment of expectations [12,35]. It is important to note that these interpretations are inferential rather than empirical, and future research should employ validated resilience scales or longitudinal designs to examine how learning disruptions affect resilience development over time.

In summary, this study provides contextualised insights into physiotherapy students' perceptions of anatomy learning during the COVID-19 pandemic. While students appreciated the continuity of online learning, they expressed a clear preference for the hands-on experience offered by the anatomy hall. Consistent with global findings, reduced interaction during online teaching highlights the need to incorporate structured engagement strategies, such as small-group tutorials, breakout discussions, or redesigned learning spaces that facilitate closer student-faculty contact, in future blended models. The results also support the continued use of online platforms for lectures and tutorials, while emphasising the importance of preserving cadaveric exposure for practical learning whenever possible. Reports of technical difficulties further underscore the need to invest in robust digital infrastructure. Future blended or resilient anatomy curricula should maintain cadaveric learning as a core component, while strategically integrating complementary digital tools such as 3D visualisation software, virtual dissection platforms, augmented reality environments, and, where feasible, haptic technologies. This balanced approach may better ensure continuity of learning during disruptions without compromising the experiential depth of face-to-face anatomy education.

Several limitations should be acknowledged. The small sample size (n = 53) from a single institution limits the generalisability of the findings, particularly as this is the only physiotherapy programme in the country. Inclusion of overseas programmes would have introduced confounding cultural, societal, and infrastructural variables. The low response rate raises the potential for non-response bias, and the results should therefore be interpreted as exploratory. The limited number of male participants precluded sex-based subgroup analyses, which should be addressed in future research with larger and more balanced samples. Data collection occurred at a single time point, preventing longitudinal analysis. Consequently, the findings are context-specific and exploratory, though they offer valuable insights to inform future multi-institutional studies. The questionnaire, while informed by previous studies and expert consensus, was not formally validated, and its heterogeneous structure precluded psychometric testing. As a result, the findings are based on self-reported perceptions, and resilience was explored indirectly rather than through a validated scale. Knowledge retention and instructional effectiveness were not directly assessed.

5. Conclusions

This study examined physiotherapy students' perceptions of anatomy education during the pandemic, revealing a clear preference for face-to-face learning, particularly for cadaveric practical sessions, alongside recognition of the value of online modalities in ensuring educational continuity. While online OSPE was perceived as less stressful, students

continued to view face-to-face assessment as more authentic and professionally relevant. The findings suggest that future anatomy curricula should adopt blended, adaptable approaches that integrate the strengths of digital tools while preserving the pedagogical and affective benefits of hands-on learning. Ensuring resilient, flexible teaching models will be essential for maintaining high-quality anatomy education during future disruptions.

Supplementary Materials: The following supporting information can be downloaded at: https://www.mdpi.com/article/10.3390/higheredu5010004/s1, Supplementary Information File S1: the survey questionnaire.

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Abbreviations

The following abbreviations are used in this manuscript:

AY19	Academic Year 2019 matriculation (September 2019 to August 2020)
AY20	Academic Year 2020 matriculation (September 2020 to August 2021)
AY21	Academic Year 2021 matriculation (September 2021 to August 2022)

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